

COMP5310 - Assignment 1

Choose a problem, prepare the data, and establish an evidence base

Key Administrative Information

- **Due date:** Sunday, 13 Sept 2026 (Week 6), 23:59 AEST
- **Weighting:** 15% of the overall unit mark (15 marks)
- **Group size:** 2 students
- **Project period:** Weeks 3-6
- **Project identifier:** Group_<YYY>_A1, where <YYY> is your group ID number in Canvas
- **Page limit:** max 10 pages (including references and cover page)
- **Cover page:** group ID number and [contribution statement table](#) (listing SID and UniKey)

Only one submission is required per group. Marks are adjusted for each group member according to the contribution statement. Raise group contribution issues at the earliest checkpoint.

Assignment Objectives

In this project, your group will act as a small data consultancy, progressing from dataset assessment and preparation to analytical modelling and evaluation.

Assignment 1 – Dataset Selection and Preparation:

Assess THREE potential client datasets, select the most promising project using evidence, and prepare the chosen dataset for further analysis in Assignment 2. This Assignment 1 focuses on evidence-based selection, transparent data cleaning, useful exploratory insight, and a reproducible handover. Predictive modelling is not required in Assignment 1.

Assignment 2 – Modelling and Evaluation:

Build on the cleaned dataset and research question developed in Assignment 1. Apply, evaluate, and compare appropriate analytical techniques to answer the research question, interpret the findings, and provide conclusions relevant to the identified stakeholder.

Project Missions and Dataset Options

- **Dataset A - NSW Train Occupancy:** advise a transport operations team about crowding patterns and where better planning or further prediction could support service decisions.
- **Dataset B - Air Travel Delay:** advise an airline or airport operations team about disruption patterns and opportunities to anticipate or reduce delays.

- **Dataset C - Hotel Booking:** advise hotel revenue or guest-service managers about booking behaviour, cancellations, demand, or operational planning.

You may propose a different stakeholder question within the selected dataset. Confirm any substantial change of scope with your tutor at the Week 4 selection checkpoint.

How the tasks in two Assignments are connected

- Audit all three datasets at a lightweight, comparable level.
- Select one dataset and complete the detailed cleaning and EDA only for that dataset.
- Carry the cleaned data, research questions, data dictionary, and limitations from Assignment 1 directly into Assignment 2.

Project Timeline

Week	Checkpoint	Evidence produced
3	Project launch & Dataset audition	Team charter, initial roles, and preferred stakeholder mission (formative); Comparison table for all three datasets.
4	Grouping DDL & Selection checkpoint	Finalize your group on Canvas; Chosen dataset, stakeholder decision, research question, and cleaning plan.
5	Evidence clinic	Draft cleaning log and three draft decision-relevant visualisations for peer feedback.
6	Final submission	Data-readiness brief, reproducible code package, cleaned dataset, and contribution statement.

Checkpoint principle: each activity creates an artefact that is refined for the final submission; checkpoints are not separate reports.

Detailed Requirements

1. Dataset Audition and Selection (2 marks)

- Create one compact comparison table covering all three datasets.
- For each dataset, report its size and variable types, a plausible outcome or analytical opportunity, major data-quality risks, a realistic stakeholder decision, and one ethical or feasibility concern.
- Select one dataset for the remainder of Assignment 1 and justify the choice using evidence from the table.
- Do not perform detailed cleaning or full EDA for the two unselected datasets.

Strong work: evidence is distinguished from preference, and the selected project is shown to be useful, feasible, and suitable for Assignment 2.

2. Stakeholder Problem and Success Criteria (2 marks)

- Identify the primary stakeholder and the concrete decision they need to make.
- State a focused research or business question and define the intended target or outcome where applicable.
- Explain what a useful answer would look like and the practical cost of a misleading conclusion.

3. Data-Readiness Pipeline (5 marks)

- Create a concise data dictionary for the variables retained for analysis.
- Audit missing values, duplicates, data types, invalid values, inconsistent categories, unusual ranges, and other dataset-specific quality issues.
- Maintain a cleaning log showing the issue, evidence, action, justification, and before/after impact.
- Implement the cleaning process as a reproducible pipeline using relative paths. Save the final cleaned dataset as a CSV file.
- Identify variables that could create target leakage or would not be available at the intended decision time.
- Justify consequential choices; listing software functions without explaining their effect is insufficient.

4. Exploratory Data Analysis (EDA) (4 marks)

- Present at least three purposeful visualisations that investigate the stakeholder question. Choose numerical, categorical, temporal, or grouped analyses as appropriate to the data.
- Interpret each result in decision-relevant language; do not merely describe chart shapes.
- Briefly outline an Assignment 2 plan based on your EDA, including key findings or limitations, the proposed target and features, suitable modelling directions, and important risks or hypotheses to investigate.
- Do not train or compare predictive models in Assignment 1.

5. Reproducibility and Team Process (2 marks)

- Notebooks must run from top to bottom in the documented order using relative paths.
- Include a README describing dependencies, file locations, run order, and expected outputs; use a fixed random seed where randomness is introduced.
- Include a concise contribution statement and evidence that roles were reviewed at the Week 6 checkpoint.

Marking Allocation

Marks	Criterion	Evidence expected
2	Dataset audition and selection	Comparable table and evidence-based selection rationale.
2	Stakeholder problem and success criteria	Decision, focused question, intended outcome, value, and consequences.
5	Data-readiness pipeline	Data dictionary, audit, justified cleaning log, leakage check, and cleaned CSV.
4	Exploratory evidence	Purposeful visual analysis, interpretation, limitations, and Assignment 2 direction.
2	Reproducibility and team process	Runnable package, README, checkpoint evidence, and contribution statement.
15	Total	Assignment 1 contributes 15% of the unit mark.

Submission Requirements

Report

- Submit one self-contained PDF with a maximum of 10 pages, including reference and cover page.
- Cover page: group number and contribution statement table (listing SID and UniKey for each member, no names)
- Include the datasets comparison table, selected stakeholder problem, cleaning evidence, EDA findings, Assignment 2 recommendation, limitations, and contribution statement.
- Filename: Group_<YYY>_A1_Report.pdf

Code and Data Package

- Submit one ZIP containing: 01_dataset_audition.ipynb, 02_data_readiness.ipynb, the cleaned CSV, and README.md.
- The second notebook must recreate the cleaned CSV from the provided raw data when run in the documented environment.
- Do not include duplicate raw datasets supplied through Canvas unless instructed.
- Filename: Group_<YYY>_A1_Code.zip

Team Contribution Process

- Agree on initial roles in Week 3 and review them at the Week 6 checkpoint. Rotate or rebalance work where needed.

- Every member should contribute to both the analytical work and quality review; dividing the report into isolated sections without integration is discouraged.
- Include the following contribution table in the final report.

Contribution agreement: All members must agree on the submitted contribution statement. Without unanimous agreement, equal contribution will be assumed. Individual marks may be adjusted where evidence demonstrates a substantial contribution imbalance.

Group Member	Main Responsibility and Contribution	Agreed Contribution (%)
Student1 SID & UniKey	...	50
Student2 SID & UniKey	...	50

Academic Integrity and Generative AI

- Follow the University's academic integrity requirements and the unit's published conditions for generative AI. Acknowledge any permitted use in the report, including the tool, purpose, and how outputs were checked.
- Your group remains responsible for the correctness, originality, and reproducibility of every submitted statement, figure, and line of code.

Mark Deduction Policy

- Late submission: A penalty of up to **5% per day** of the maximum assignment mark applies to late submission of the report or code package. Submissions with more than **3 calendar days late** receive a mark of zero.
- Non-compliance with guidelines: A penalty of **up to 15%** of the maximum assignment mark applies to submissions that fail to follow formatting or administrative guidelines. Common examples include, but are not limited to: Failing to form groups properly on Canvas; Omitting student identification details (SID or Unikey) on the cover page, etc.